

Google Cybersecurity Professional Certificate

Wireshark DNS Query Portfolio Activity

This portfolio activity documents the practical analysis of Domain Name System (DNS) traffic using Wireshark as part of the **Google Cybersecurity Professional Certificate** program. The objective of this activity was to inspect DNS query packets, identify queried domains, and understand how DNS resolution processes function within network communication.

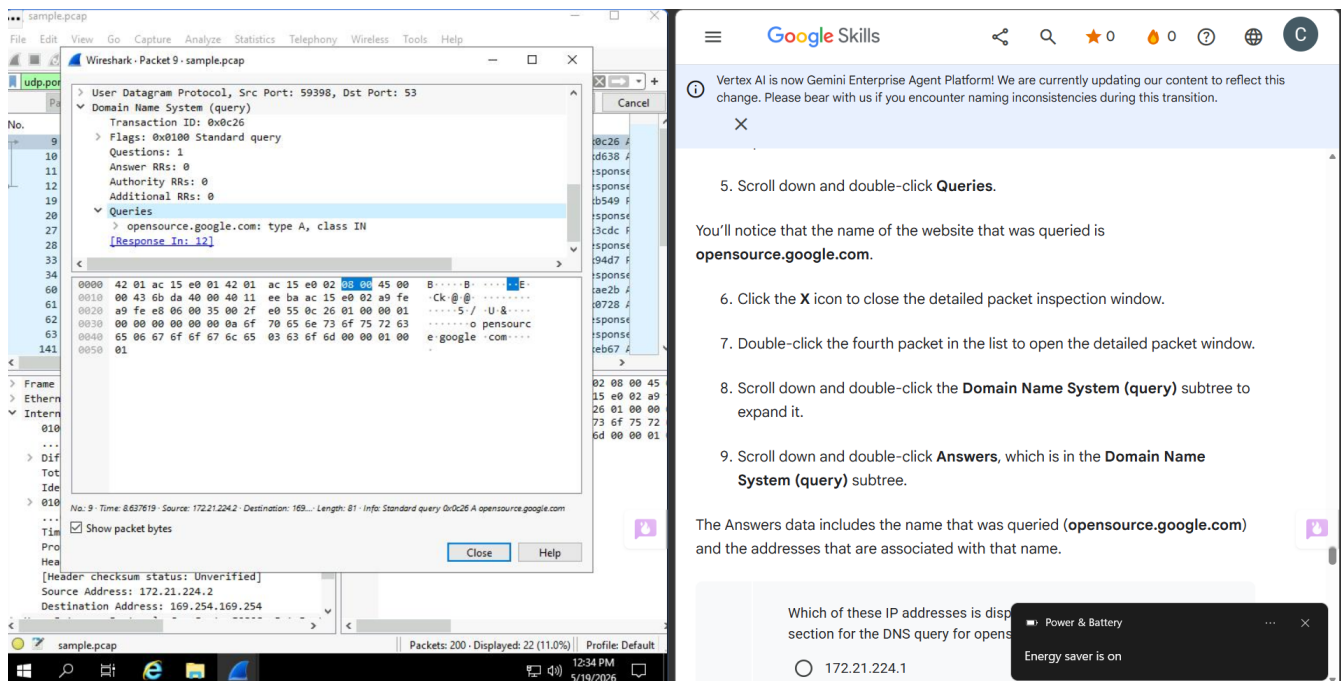
Activity Objectives

1. Capture and inspect DNS query traffic using Wireshark.
2. Identify queried domain names and associated packet details.
3. Analyze DNS request structures and protocol behavior.
4. Develop foundational network traffic analysis skills for cybersecurity investigations.

Methodology and Procedure

During the activity, Wireshark was used to inspect packets from a sample packet capture (PCAP) file. The DNS query packet was opened and expanded to examine the **Domain Name System (query)** subtree. The analysis focused on identifying the queried domain, transaction details, and packet structure. The DNS query revealed that the queried domain was: **opensource.google.com** The packet inspection also showed the following information: Protocol: UDP Destination Port: 53 (DNS) Query Type: A Record Query Class: IN (Internet)

Traffic Analysis Screenshot



Key Findings

The DNS query packet analysis confirmed successful inspection of DNS request traffic within Wireshark. The captured packet demonstrated how a client system communicates with a DNS server to resolve a domain name into an IP address. Understanding these packet structures is essential in cybersecurity operations, especially in network monitoring, threat detection, incident response, and forensic analysis. This exercise also reinforced the importance of packet analysis tools such as Wireshark in identifying network communication patterns and troubleshooting connectivity issues.

Conclusion

This portfolio activity provided practical experience in analyzing DNS traffic using Wireshark within the Google Cybersecurity Professional Certificate program. The activity enhanced understanding of DNS query operations and strengthened foundational cybersecurity analysis skills relevant to network defense and security monitoring environments.

Prepared as part of the Google Cybersecurity Professional Certificate coursework.